

Homework 1

Due : Friday, February 20 at 11 : 59 pm

Deliverables. Submit a single PDF of your write-up to Gradescope *HW1 Write-Up*.

Start each problem on a new page.

Honor Code

Write and sign the following statement:

“I certify that all solutions in this document are entirely my own and that I have not looked at anyone else’s solution. I have given credit to all external sources I consulted.”

Jingliang Tu

Linear Algebra

1. **System of equations (3 points).** Consider the linear system

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 6 \\ 3 & 6 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \\ 12 \end{bmatrix}.$$

How many solutions does this system have? Explain your reasoning.

Answer: 0 solutions.

Reasoning: Let us represent the system as an augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 2 & 0 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{array} \right]$$

Notice that the third row of the coefficient matrix is exactly 3 times the first row ($3 \times [1, 2, 3] = [3, 6, 9]$). However, for the right-hand side, multiplying the first row’s value by 3 yields $3 \times 6 = 18$, but the third equation specifies 12. This implies $3x + 6y + 9z = 18$ and $3x + 6y + 9z = 12$, which is a contradiction ($18 \neq 12$). Therefore, the system is inconsistent and has no solutions.

2. **Asymptotic powers of 2×2 matrices (6 points).** For each matrix below, determine the behavior of

$$\lim_{n \rightarrow \infty} M^n$$

Hint: Use the eigen-decomposition $M = PDP^{-1}$, where $D = \{\lambda_1, \lambda_2, \dots\}$ is a diagonal matrix of the eigenvalues. What would be the formula for M^n ?

(a) $M_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$

Answer: The limit does not exist.

Reasoning: The characteristic equation for M_1 is $\det(M_1 - \lambda I) = \lambda^2 - 1 = 0$, giving eigenvalues $\lambda_1 = 1$ and $\lambda_2 = -1$. Since $M_1^n = PD^nP^{-1}$ and $D^n = \begin{bmatrix} 1^n & 0 \\ 0 & (-1)^n \end{bmatrix}$, the term $(-1)^n$ alternates between 1 and -1 . Thus, the sequence of matrices oscillates between the identity matrix and M_1 , and does not converge.

(b) $M_2 = \begin{bmatrix} 2 & -5 \\ 1/2 & -7/6 \end{bmatrix}.$

Answer: The limit is the zero matrix $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$

Reasoning: The trace of M_2 is $2 - 7/6 = 5/6$, and the determinant is $2(-7/6) - (-5)(1/2) = -14/6 + 5/2 = 1/6$. The characteristic equation is $\lambda^2 - \frac{5}{6}\lambda + \frac{1}{6} = 0$, which factors as $(\lambda - 1/2)(\lambda - 1/3) = 0$. The eigenvalues are $\lambda_1 = 1/2$ and $\lambda_2 = 1/3$. Since $|\lambda_i| < 1$ for both eigenvalues, $\lim_{n \rightarrow \infty} \lambda_i^n = 0$. Therefore, $\lim_{n \rightarrow \infty} M_2^n = P \lim_{n \rightarrow \infty} D^n P^{-1} = P \mathbf{0} P^{-1} = \mathbf{0}$.

(c) $M_3 = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}.$

Answer: The limit diverges (does not exist).

Reasoning: The trace is $4 + 1 = 5$, and the determinant is $4(1) - (-2)(1) = 6$. The characteristic equation is $\lambda^2 - 5\lambda + 6 = 0$, factoring as $(\lambda - 2)(\lambda - 3) = 0$. The eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = 3$. Since there are eigenvalues with absolute value strictly greater than 1, the entries of D^n will grow without bound to infinity as $n \rightarrow \infty$, causing the matrix powers to diverge.

3. **Singular Value Decomposition (4 points).** Given the matrix

$$A = \begin{bmatrix} 2 & -1 \\ 2 & 2 \end{bmatrix}$$

Find a unit vector x ($\|x\| = 1$) for which $\|Ax\|$ is maximized. *Hint: Recall from SVD that $A = U\Sigma V^\top$, where A maps each right singular vector (a column of V) to a left singular vector (a column of U), scaled by the corresponding singular value. For which right singular vector does this scaling make $\|Ax\|$ the largest?*

Answer: $x = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}$ (or its negation).

Reasoning: To maximize $\|Ax\|$ subject to $\|x\| = 1$, we need to find the right singular vector corresponding to the largest singular value of A . This is equivalent to finding the eigenvector corresponding to the largest eigenvalue of $A^\top A$.

$$A^\top A = \begin{bmatrix} 2 & 2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 8 & 2 \\ 2 & 5 \end{bmatrix}$$

The characteristic equation is $\det(A^\top A - \lambda I) = (8 - \lambda)(5 - \lambda) - 4 = \lambda^2 - 13\lambda + 36 = 0$. Factoring gives $(\lambda - 9)(\lambda - 4) = 0$, so the eigenvalues are $\lambda_1 = 9$ and $\lambda_2 = 4$. The largest eigenvalue is $\lambda_1 = 9$. We find its corresponding eigenvector by solving $(A^\top A - 9I)v = 0$:

$$\begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies -v_1 + 2v_2 = 0 \implies v_1 = 2v_2$$

A vector in this eigenspace is $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Normalizing it to be a unit vector gives $x = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}$.

4. **Image Flipping (6 points).** *This question is a warmup to problem 4 in the coding section.* Consider a tiny 3×3 image represented by matrix I and the same image flipped on the Y axis I_{flip} :

$$I = \begin{bmatrix} x_1 & x_2 & x_3 \\ x_4 & x_5 & x_6 \\ x_7 & x_8 & x_9 \end{bmatrix}, \quad I_{\text{flip}} = \begin{bmatrix} x_7 & x_8 & x_9 \\ x_4 & x_5 & x_6 \\ x_1 & x_2 & x_3 \end{bmatrix}.$$

- (a) What is the size of the transformation matrix T that performs this flip? *Hint: You can first convert I to a 1×9 vector, make transformation using matrix T , and then convert I_{flip} back to a 3×3 matrix.*

Answer: 9×9 .

Since I is flattened into a 1×9 vector, and we want to multiply it by T to get another 1×9 vector, the transformation matrix T must have dimensions 9×9 .

- (b) Construct T and verify that $I \times T$ produces I_{flip} .

Answer: Let the flattened vector be $I_{\text{vec}} = [x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9]$. We want $I_{\text{vec}}T = [x_7, x_8, x_9, x_4, x_5, x_6, x_1, x_2, x_3]$. The matrix T that swaps the first and third blocks of size 3 is:

$$T = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Multiplying $I_{\text{vec}} \times T$ correctly routes x_1, x_2, x_3 to the last three positions and x_7, x_8, x_9 to the first three positions, matching I_{flip} when reshaped.

- (c) Describe an algorithm for constructing the vertical-flip transformation matrix for any $N \times N$ matrix (either text explanation or pseudocode is acceptable). Show that $I \times T$ produces I_{flip} .

Answer: Algorithm for vertical flip of an $N \times N$ matrix:

- i. Flatten the $N \times N$ matrix row by row into a $1 \times N^2$ vector.
- ii. Initialize an $N^2 \times N^2$ matrix T with all zeros.
- iii. For each index $i \in [0, N^2 - 1]$ in the flattened vector:
 - A. Calculate its original row $r = \lfloor i/N \rfloor$ and column $c = i \bmod N$.
 - B. The new row after a vertical flip is $r' = N - 1 - r$. The column remains c .
 - C. Calculate the new flattened index $i' = r' \times N + c$.
 - D. Set $T_{i+1, i'+1} = 1$ (using 1-based indexing for matrices).
- iv. Multiply the flattened vector by T and reshape the result back to $N \times N$.

By construction, T is a permutation matrix that maps the element at (r, c) to $(N - 1 - r, c)$. This accurately performs the row reversal seen in a vertical flip.

- (d) Describe how you would modify this algorithm to do a *horizontal* flip (along the X axis).

Answer: To perform a horizontal flip, we modify the step that calculates the new row and column. Instead of changing the row, we change the column index. For each original

position (r, c) , the new position is (r', c') where $r' = r$ and $c' = N - 1 - c$. The new flattened index is $i' = r \times N + c'$. We then set $T_{i+1, i'+1} = 1$. This swaps the columns left-to-right rather than the rows top-to-bottom.

Calculus

5. Partial Derivatives (8 points). (First and second derivatives)

- (a) For $f(x_1, x_2) = x_1^3 + x_2^3 - 3x_1x_2$, find all the first and second order partial derivatives.

First-order derivatives:

$$\frac{\partial f}{\partial x_1} = 3x_1^2 - 3x_2$$

$$\frac{\partial f}{\partial x_2} = 3x_2^2 - 3x_1$$

Second-order derivatives:

$$\frac{\partial^2 f}{\partial x_1^2} = 6x_1$$

$$\frac{\partial^2 f}{\partial x_2^2} = 6x_2$$

$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = \frac{\partial^2 f}{\partial x_2 \partial x_1} = -3$$

- (b) Let $f(x, y) = 4x^2 + y^2 - 8xy + 4x + 6y - 10$. Find the critical points by solving

$$\frac{\partial f}{\partial x} = 0 \quad \text{and} \quad \frac{\partial f}{\partial y} = 0$$

simultaneously and determine which point(s) yield a minimum value.

Answer: We solve the system of equations:

$$\frac{\partial f}{\partial x} = 8x - 8y + 4 = 0 \implies x - y = -0.5$$

$$\frac{\partial f}{\partial y} = 2y - 8x + 6 = 0 \implies -4x + y = -3$$

Adding the two equations: $(x - y) + (-4x + y) = -0.5 - 3 \implies -3x = -3.5 \implies x = \frac{7}{6}$.
Substituting x back: $\frac{7}{6} - y = -\frac{1}{2} \implies y = \frac{7}{6} + \frac{3}{6} = \frac{10}{6} = \frac{5}{3}$. The only critical point is $(\frac{7}{6}, \frac{5}{3})$.

To determine if it is a minimum, we use the second derivative test. The Hessian matrix is:

$$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} = \begin{bmatrix} 8 & -8 \\ -8 & 2 \end{bmatrix}$$

The determinant of the Hessian is $D = (8)(2) - (-8)^2 = 16 - 64 = -48$. Since $D < 0$, the critical point is a **saddle point**. Therefore, this function **does not have a minimum value**.

- (c) For $f(x, y) = e^{xy} + x^2y$, compute:

- (i) $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$

$$\frac{\partial f}{\partial x} = ye^{xy} + 2xy$$

$$\frac{\partial f}{\partial y} = xe^{xy} + x^2$$

(ii) $\frac{\partial^2 f}{\partial x^2}$, $\frac{\partial^2 f}{\partial y^2}$, and $\frac{\partial^2 f}{\partial x \partial y}$

$$\frac{\partial^2 f}{\partial x^2} = y^2 e^{xy} + 2y$$

$$\frac{\partial^2 f}{\partial y^2} = x^2 e^{xy}$$

$$\frac{\partial^2 f}{\partial x \partial y} = e^{xy} + xy e^{xy} + 2x = (xy + 1)e^{xy} + 2x$$

(d) For $f(x, y) = \ln(x^2 + y^2 + 1)$, compute:

(i) $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$

$$\frac{\partial f}{\partial x} = \frac{2x}{x^2 + y^2 + 1}$$

$$\frac{\partial f}{\partial y} = \frac{2y}{x^2 + y^2 + 1}$$

(ii) $\frac{\partial^2 f}{\partial x^2}$, $\frac{\partial^2 f}{\partial y^2}$, and $\frac{\partial^2 f}{\partial x \partial y}$

$$\frac{\partial^2 f}{\partial x^2} = \frac{2(x^2 + y^2 + 1) - 2x(2x)}{(x^2 + y^2 + 1)^2} = \frac{-2x^2 + 2y^2 + 2}{(x^2 + y^2 + 1)^2}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{2(x^2 + y^2 + 1) - 2y(2y)}{(x^2 + y^2 + 1)^2} = \frac{2x^2 - 2y^2 + 2}{(x^2 + y^2 + 1)^2}$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{2y}{x^2 + y^2 + 1} \right) = \frac{0 - 2y(2x)}{(x^2 + y^2 + 1)^2} = \frac{-4xy}{(x^2 + y^2 + 1)^2}$$

6. **Recursive expression and derivatives (6 points).** Suppose we have variables $z_1, z_2, \dots, z_n$, $w_1, \dots, w_{n-1}$, and $b_1, \dots, b_{n-1} \in \mathbb{R}$, where

$$z_n = w_{n-1}z_{n-1} + b_{n-1}.$$

- (a) Compute $\frac{dz_k}{dz_{k-1}}$ for $k = 2, \dots, n$.

$$\frac{dz_k}{dz_{k-1}} = w_{k-1}$$

- (b) Compute $\frac{dz_n}{dz_1}$.

By the chain rule:

$$\frac{dz_n}{dz_1} = \frac{dz_n}{dz_{n-1}} \cdot \frac{dz_{n-1}}{dz_{n-2}} \cdots \frac{dz_2}{dz_1} = \prod_{i=1}^{n-1} w_i$$

- (c) Compute $\frac{dz_n}{db_1}$. You may express your answers in terms of $z_1, \dots, z_n, w_1, \dots, w_{n-1}, b_1, \dots, b_{n-1}$

Notice that b_1 only directly affects z_2 via the equation $z_2 = w_1z_1 + b_1$. Thus, the partial derivative $\frac{\partial z_2}{\partial b_1} = 1$. Applying the chain rule from z_n down to z_2 :

$$\frac{dz_n}{db_1} = \frac{dz_n}{dz_2} \cdot \frac{\partial z_2}{\partial b_1} = \left(\prod_{i=2}^{n-1} w_i \right) \cdot 1 = \prod_{i=2}^{n-1} w_i$$

Probability

7. **Conditioned uniform difference (3 points).** Let $X, Y \stackrel{\text{iid}}{\sim} \text{Unif}(-1, 1)$. Compute

$$\mathbb{P}(|X - Y| \leq 0.5 \mid XY > 0).$$

Hint: Try drawing a 2D cartesian plane where the horizontal and vertical axes represent X and Y respectively and each range from -1 to 1. For which quadrants is it true that $XY > 0$? Within these quadrants, how can we visualize the region $|X - Y| < 0.5$?

Answer: $3/4$.

Reasoning: The sample space for (X, Y) is the square $[-1, 1] \times [-1, 1]$ with total area 4. The condition $XY > 0$ restricts our space to the first and third quadrants (where X and Y have the same sign). The total area of this conditioned space is $1 + 1 = 2$. Within the first quadrant ($0 < X, Y < 1$), the region $|X - Y| \leq 0.5$ is the unit square minus two triangles at the corners (one where $X - Y > 0.5$, and one where $Y - X > 0.5$). Each triangle has a base of 0.5 and a height of 0.5, so the area of both triangles combined is $2 \times \frac{1}{2}(0.5)(0.5) = 0.25$. Thus, the area satisfying the condition in the first quadrant is $1 - 0.25 = 0.75$. By symmetry, the valid area in the third quadrant is also 0.75. The total favorable area is $0.75 + 0.75 = 1.5$. The conditional probability is the favorable area divided by the total conditioned area: $\frac{1.5}{2} = \frac{3}{4} = 0.75$.

8. **Nearest-neighbor arc length (4 points).** Suppose you select 20 i.i.d. points $X_1, \dots, X_{20}$ uniformly at random on the circumference of the unit circle.

- (a) Let D be the shortest arc distance from X_1 to the nearest of the other 19 points. Calculate $\mathbb{P}(D > t)$ where $0 \leq t \leq \frac{1}{2}$. *Hint: What does the event $\{D > t\}$ mean in terms of where the other 19 points can be located relative to X_1 ?*

Answer: $\mathbb{P}(D > t) = (1 - 2t)^{19}$

Reasoning: The event $\{D > t\}$ means that none of the other 19 points fall within an arc distance of t from X_1 . For a single point, the "forbidden" region around X_1 consists of an arc of length t in both directions, totaling a length of $2t$. Since the circumference of the unit circle is normalized to 1 (representing probability), the probability that a single point falls outside this region is $1 - 2t$. Because the 19 points are selected independently and uniformly, the probability that all 19 fall outside this region is $(1 - 2t)^{19}$.

- (b) Find the expected arc length (in degrees) between X_1 and the point nearest to it. *Hint: The Wikipedia page on expected value of continuous variables may be helpful here.*

Answer: 9°

Reasoning: Since D is a continuous, non-negative random variable bounded by $1/2$, its expected value can be computed by integrating its survival function:

$$\mathbb{E}[D] = \int_0^{1/2} \mathbb{P}(D > t) dt = \int_0^{1/2} (1 - 2t)^{19} dt$$

Let $u = 1 - 2t$, then $du = -2dt$, and the limits change from $[0, 1/2]$ to $[1, 0]$.

$$\mathbb{E}[D] = \int_1^0 u^{19} \left(-\frac{1}{2} du\right) = \frac{1}{2} \int_0^1 u^{19} du = \frac{1}{2} \left[\frac{u^{20}}{20} \right]_0^1 = \frac{1}{40}$$

This means the expected arc length is $\frac{1}{40}$ of the full circle. In degrees, this is $\frac{1}{40} \times 360^\circ = 9^\circ$.

9. **Cancer screening (3 points).** A medical test has sensitivity 90% and false-positive rate 3%:

$$\mathbb{P}(T = 1 \mid C = 1) = 0.9, \quad \mathbb{P}(T = 1 \mid C = 0) = 0.03.$$

Suppose the disease prevalence is very low, $\mathbb{P}(C = 1) = 0.001$. Compute the posterior probability of disease given a positive test:

$$\mathbb{P}(C = 1 \mid T = 1).$$

Answer: Approximately 0.0291 or 2.91%.

Reasoning: By Bayes' Theorem:

$$\mathbb{P}(C = 1 \mid T = 1) = \frac{\mathbb{P}(T = 1 \mid C = 1)\mathbb{P}(C = 1)}{\mathbb{P}(T = 1)}$$

First, we find the marginal probability of a positive test, $\mathbb{P}(T = 1)$, using the Law of Total Probability:

$$\begin{aligned} \mathbb{P}(T = 1) &= \mathbb{P}(T = 1 \mid C = 1)\mathbb{P}(C = 1) + \mathbb{P}(T = 1 \mid C = 0)\mathbb{P}(C = 0) \\ &= (0.9)(0.001) + (0.03)(1 - 0.001) \\ &= 0.0009 + (0.03)(0.999) \\ &= 0.0009 + 0.02997 = 0.03087 \end{aligned}$$

Now, apply Bayes' Theorem:

$$\mathbb{P}(C = 1 \mid T = 1) = \frac{0.0009}{0.03087} \approx 0.02915$$

10. **Follower Counts (3 points).** Suppose 420 people are sitting uniformly at random around a circle, each with a *distinct* number of TikTok followers. What is the expected number of people whose follower count is higher than both their immediate neighbors?

Answer: 140

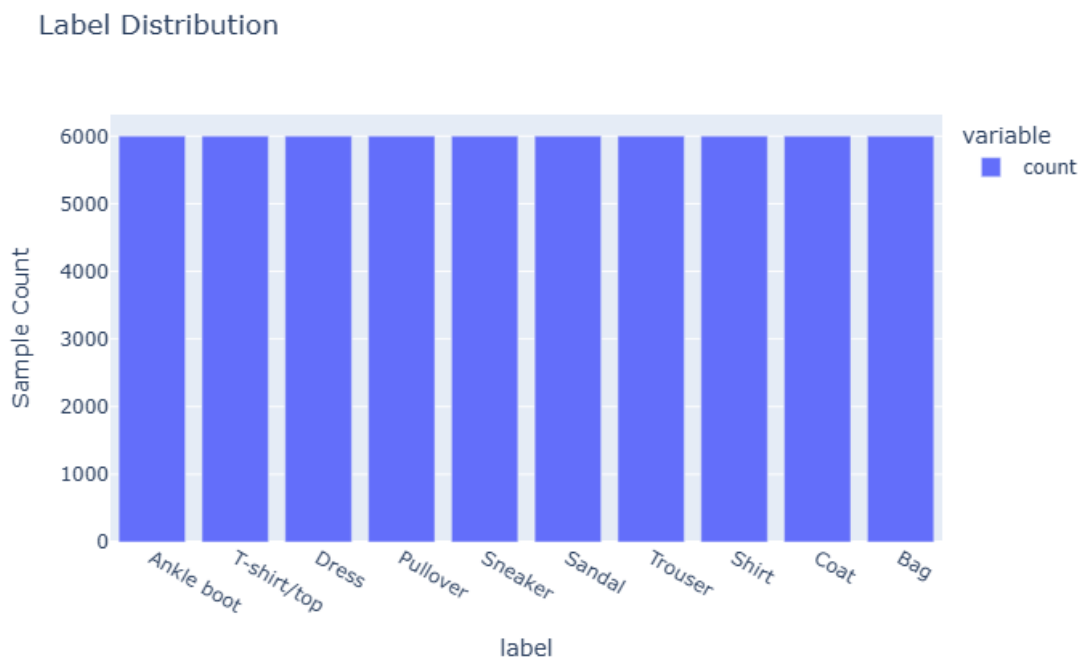
Reasoning: Let I_i be an indicator variable that is 1 if person i has more followers than both of their immediate neighbors, and 0 otherwise. For any group of 3 adjacent people, all have distinct follower counts. The probability that the middle person has the highest count among these 3 is exactly $\frac{1}{3}$, by symmetry. Thus, $\mathbb{E}[I_i] = \frac{1}{3}$. Let $X = \sum_{i=1}^{420} I_i$ be the total number of such people. Using the linearity of expectation:

$$\mathbb{E}[X] = \sum_{i=1}^{420} \mathbb{E}[I_i] = 420 \times \frac{1}{3} = 140$$

Coding

For every question labeled with **MANUAL** in the Coding part of this homework, please label and attach either your figures or your written answers below.

1c: Visualizing Label Distribution



2c: Visualizing Clusters

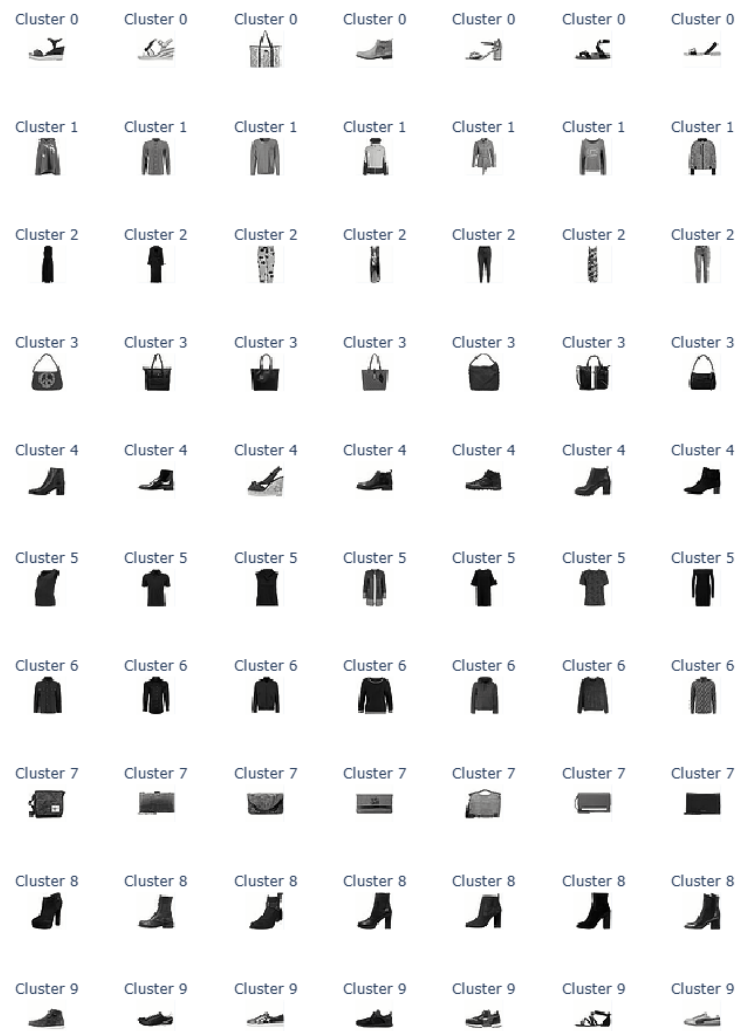


Figure 1: I resized the image for better displaying

4g: Matrix Multiply Questions

1. Does the order in which you apply transformations matter? Why or why not?

Answer: Yes, the order absolutely matters. In linear algebra, matrix multiplication is generally not commutative ($T_1T_2 \neq T_2T_1$). Geometrically, applying a translation (shift) and then rotating the image around the origin will yield a completely different result compared to rotating the image first and then translating it.

2. When can a transformation be undone (i.e., when can you multiply your augmented image by another transformation matrix to recover the original image)? What matrix would you multiply by to recover the original image?

Answer: A transformation can be undone if and only if its corresponding transformation matrix T is **invertible** (i.e., it is a non-singular matrix with a non-zero determinant). To recover the original image, you would multiply the augmented image by the **inverse matrix**, T^{-1} (specifically, applying $(T^{-1})^\top$ to the flattened image vector).

3. Which of the augmentations implemented above can be "undone"? For augmentations that can be undone but may lose information (e.g., parts of the image are cut off), explain the conditions under which this occurs.

Answer:

- **Flipping (Horizontal/Vertical):** Can be perfectly undone. The transformation matrix is its own inverse, so applying the exact same flip again restores the original image perfectly without any information loss.
- **Shifting:** Can technically be undone by shifting in the opposite direction (e.g., multiplying by a shift matrix with $-dx, -dy$). **However, information is lost** if the initial shift pushes any pixels outside the 28×28 image boundaries. Those pixels are discarded and replaced by 0s, meaning the opposite shift will just bring back black pixels instead of the original edge content. It can only be perfectly undone if the shifted content remains entirely within the image bounds.
- **Rotation (multiples of 90°):** Can be perfectly undone by rotating in the opposite direction. (Other discrete rotations suffer from rounding/interpolation loss and cannot be perfectly inverted).

4. Which of these augmentations *cannot* be "undone" with another matrix multiplication? Why not?

Answer: **Blurring** cannot be undone. Mathematically, the blurring matrix averages neighboring pixels together, which acts as a low-pass filter. This matrix is singular (or highly ill-conditioned), meaning its inverse does not exist. Intuitively, the blurring process destroys high-frequency details; multiple different sharp images could potentially result in the exact same blurred image. Because the mapping is not one-to-one, we cannot uniquely reconstruct the original pixel values via matrix multiplication. Furthermore, general **Rotations (non- 90°)** using nearest-neighbor or bilinear interpolation also cannot be perfectly undone due to the loss of information from pixel quantization (rounding coordinates) and boundary cropping.

4j: Analysis of Augmentation Techniques

Based on the results shown in the bar chart, the performance varies significantly depending on the type of augmentation applied.

Best Performing Augmentations: The blurring augmentations performed the best, specifically `blur_3x3` (accuracy of 0.89) and `blur_5x5` (accuracy of 0.73). **Reasoning:** Blurring acts as a low-pass filter that softens edges but preserves the global spatial structure, silhouette, and center of mass of the clothing items. Because the Multi-Layer Perceptron (MLP) model expects specific shapes to be in specific pixel locations, keeping the main body of the object in its original place allows the model to successfully recognize the item despite the loss of fine details.

Worst Performing Augmentations: The composite transformations and severe rotations performed the worst. For instance, `shift_5_0_rotate_45_blur_3x3` yielded an accuracy of just 0.03, and `rotate_90` yielded 0.06. **Reasoning:** MLPs lack spatial invariance (unlike Convolutional Neural Networks). Each pixel index is treated as an independent feature with its own learned weight. When an image is rotated by 90° or drastically shifted, the defining features of an object (e.g., the vertical leg of a trouser or the handle of a bag) are mapped to entirely different pixel locations. The model perceives this displaced structure as completely unseen noise, destroying its ability to generalize and resulting in accuracy that is close to random guessing.

5c: MAE, MSE, and R-Squared

1. Advantages of MAE vs. MSE:

- **MAE over MSE:** MAE is more robust to outliers because it treats all errors linearly. MSE squares the errors, meaning a single large outlier can disproportionately affect the loss, potentially biasing the model.
- **MSE over MAE:** MSE is mathematically easier to optimize (differentiable at zero) and is more sensitive to large errors, which is beneficial if small errors are acceptable but large deviations are catastrophic.

- ### 2. Significance of R-squared: Answer:
- R-squared (R^2) measures the proportion of variance in the dependent variable that is explained by the independent variables in the model. It is significant because it provides a normalized measure of how well the model fits the data compared to a baseline model that always predicts the mean value.

6d: Analysis of Class Accuracies and Confusion Matrix

Answer: The data visualizations reveal a severe and systematic degradation in model performance. According to the class-wise accuracy plot, the accuracy for the secret test set has plummeted across all categories, with most falling below 20%.

The confusion matrix highlights a highly unusual pattern: a vast number of images from diverse categories—such as **Pullover** (111 instances), **Ankle boot** (99), and **Shirt** (77)—are consistently misclassified as **Bag**. This indicates that the misclassification is not random noise but a systematic error. Such a pattern suggests a major distribution shift in the test set’s spatial characteristics. Specifically, the defining geometric features of these items have been moved to pixel locations that the MLP previously associated with the rectangular or boxy shape of a **Bag**. As noted in the following sections of the notebook, this is a hallmark of global image rotation, which completely misaligns the absolute pixel positions that the MLP relies on for inference.

8d: Comparing Data Augmentation and Rotation Correction

Based on the experimental results shown in the comparison bar chart, Solution 1 (Data Augmentation) significantly outperforms Solution 2 (Rotation Correction), achieving an accuracy of approximately 0.67 compared to 0.43. There are several key reasons why the "predict and unrotate" approach is less effective:

- **Error Propagation:** Solution 2 is a multi-step pipeline where the final classification depends entirely on the precision of the `MLPRegressor`. If the regression model predicts the angle with even a small margin of error (e.g., a few degrees), the resulting "unrotated" image remains spatially misaligned. Since MLPs lack spatial invariance, even a slight misalignment causes features to fall into the wrong pixel indices, leading to misclassification.
- **Interpolation Artifacts and Information Loss:** The process of rotating an image digitally involves pixel interpolation (e.g., bilinear or nearest-neighbor). Performing this twice—once when the secret test set was generated and once during our correction—introduces blurring, aliasing, and loss of fine details. These artifacts make the recovered image "noisier" than the sharp images the original model was trained on.
- **Intrinsic Invariance vs. Pre-processing:** Solution 1 forces the model to learn "rotation-invariant" features (features that characterize the object regardless of its orientation). This makes the model inherently more robust. Solution 2, however, tries to force the input data to fit a rigid, orientation-sensitive model, which is a much more brittle strategy in the presence of noise.